

Solving Linear Systems by Elimination

Answer Key

Algebra 1

Quick Answers

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|--|---|
| 1. $x = 4, y = 2$ | 6. $f = 5, n = 3$ |
| 2. $x = 3, y = 2$ | 7. $x = 1, y = 3$ |
| 3. $x = 3, y = 4$ | 8. (a) price per box = $p = 6, q = 8, —$ |
| 4. (a) ticket prices = $a = 9, s = 5, —$ | 9. $x = 3, y = -3$ |
| 5. $x = 3, y = 2$ | 10. (a) price per kilogram = $c = 13, d = 9, —$ |
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What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-REI.C.6 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 13 tagged checks.

- 1.** Solve by elimination: $3x + 2y = 16$ and $5x - 4y = 12$.

Solution: The y -terms are $+2y$ and $-4y$. Multiply the first equation by 2 so they become exact opposites.

$$6x + 4y = 32 \quad \text{and} \quad 5x - 4y = 12$$

Add the two equations: $11x = 44$, so $x = 4$. Substitute into $3x + 2y = 16$: $12 + 2y = 16$, so $y = 2$. Check in the second equation: $5(4) - 4(2) = 20 - 8 = 12$ ✓

$x = 4, y = 2$

- 2.** Solve by elimination: $4x + 3y = 18$ and $2x - y = 4$.

Solution: Multiply the second equation by 3 so its y -term becomes $-3y$, the opposite of $+3y$.

$$4x + 3y = 18 \quad \text{and} \quad 6x - 3y = 12$$

Add: $10x = 30$, so $x = 3$. Substitute into $2x - y = 4$: $6 - y = 4$, so $y = 2$. Check in the first equation: $4(3) + 3(2) = 18$ ✓

$x = 3, y = 2$

3. Solve by elimination: $3x + 4y = 25$ and $2x + 3y = 18$.

Solution: No coefficient of one equation divides the matching coefficient of the other, so scale both. The least common multiple of 3 and 2 is 6: multiply the first equation by 2 and the second by 3.

$$6x + 8y = 50 \quad \text{and} \quad 6x + 9y = 54$$

Subtract the first scaled equation from the second: $y = 4$. Substitute into $3x + 4y = 25$: $3x + 16 = 25$, so $x = 3$. Check in the second original equation: $2(3) + 3(4) = 18$ ✓

$x = 3, y = 4$

4. Drama club tickets. Three adult and two student tickets cost \$37; five adult and four student tickets cost \$65. (a) Find both prices in dollars. (b) Explain why one multiplication is enough to eliminate s .

Solution (a): With a the adult price and s the student price, both in dollars,

$$3a + 2s = 37 \quad \text{and} \quad 5a + 4s = 65.$$

Multiply the first equation by 2: $6a + 4s = 74$. Subtract the second equation: $a = 9$. Substitute into $3a + 2s = 37$: $27 + 2s = 37$, so $s = 5$. Check: $5(9) + 4(5) = 45 + 20 = 65$ ✓

$a = 9, s = 5$

(b) — **instructor-judged.** A correct response says that doubling the first equation turns $2s$ into $4s$, which already matches the $4s$ in the second equation, so subtracting removes s and leaves a single equation in a . Full credit for naming the matched s -coefficients as the reason; partial credit for saying only that the numbers were made the same without naming which variable is removed.

5. Solve by elimination: $5x + 3y = 21$ and $2x + 5y = 16$.

Solution: Match the x -terms. The least common multiple of 5 and 2 is 10: multiply the first equation by 2 and the second by 5.

$$10x + 6y = 42 \quad \text{and} \quad 10x + 25y = 80$$

Subtract: $19y = 38$, so $y = 2$. Substitute into $5x + 3y = 21$: $5x + 6 = 21$, so $x = 3$. Check in the second original equation: $2(3) + 5(2) = 16$ ✓

$x = 3, y = 2$

6. Office order. Four notebooks and three folders cost \$27; six notebooks and five folders cost \$43. Find the price of one notebook and one folder, in dollars.

Solution: With n the notebook price and f the folder price, both in dollars,

$$4n + 3f = 27 \quad \text{and} \quad 6n + 5f = 43.$$

Match the n -terms: the least common multiple of 4 and 6 is 12, so multiply the first equation by 3 and the second by 2.

$$12n + 9f = 81 \quad \text{and} \quad 12n + 10f = 86$$

Subtract: $f = 5$. Substitute into $4n + 3f = 27$: $4n + 15 = 27$, so $n = 3$. Check: $6(3) + 5(5) = 18 + 25 = 43$ ✓

$n = 3, f = 5$

7. Solve by elimination: $7x - 2y = 1$ and $3x + 4y = 15$.

Solution: The y -terms are $-2y$ and $+4y$. Multiply the first equation by 2 so they become opposites.

$$14x - 4y = 2 \quad \text{and} \quad 3x + 4y = 15$$

Add: $17x = 17$, so $x = 1$. Substitute into $3x + 4y = 15$: $3 + 4y = 15$, so $y = 3$. Check in the first original equation: $7(1) - 2(3) = 1$ ✓

$x = 1, y = 3$

8. Pencil and pen boxes. Five boxes of pencils and two boxes of pens cost \$46; three boxes of pencils and four boxes of pens cost \$50. (a) Find both prices in dollars. (b) Explain why scaling an equation leaves the solution set unchanged.

Solution (a): With p the price of a box of pencils and q the price of a box of pens, both in dollars,

$$5p + 2q = 46 \quad \text{and} \quad 3p + 4q = 50.$$

Multiply the first equation by 2: $10p + 4q = 92$. Subtract the second equation: $7p = 42$, so $p = 6$. Substitute into $5p + 2q = 46$: $30 + 2q = 46$, so $q = 8$. Check: $3(6) + 4(8) = 18 + 32 = 50$ ✓

$p = 6, q = 8$

(b) — instructor-judged. A correct response says that multiplying *both sides* of an equation by the same nonzero number produces an equivalent equation: every pair (p, q) that made the original true still makes the scaled one true, and the step can be undone by dividing. Full credit for tying the equivalence to multiplying both sides; no credit for answering only that the numbers get bigger.

9. Solve by elimination: $6x + 5y = 3$ and $8x - 3y = 33$.

Solution: Match the y -terms. The least common multiple of 5 and 3 is 15: multiply the first equation by 3 and the second by 5.

$$18x + 15y = 9 \quad \text{and} \quad 40x - 15y = 165$$

Add: $58x = 174$, so $x = 3$. Substitute into $6x + 5y = 3$: $18 + 5y = 3$, so $5y = -15$ and $y = -3$. Check in the second original equation: $8(3) - 3(-3) = 24 + 9 = 33$ ✓

$x = 3, y = -3$

10. Cashews and almonds. Three kilograms of cashews and two kilograms of almonds cost \$57; five kilograms of cashews and four kilograms of almonds cost \$101. (a) Find both prices per kilogram in dollars. (b) Compare the two scaling routes.

Solution (a): With c the cashew price and d the almond price, both dollars per kilogram,

$$3c + 2d = 57 \quad \text{and} \quad 5c + 4d = 101.$$

Multiply the first equation by 2: $6c + 4d = 114$. Subtract the second equation: $c = 13$. Substitute into $3c + 2d = 57$: $39 + 2d = 57$, so $d = 9$. Check: $5(13) + 4(9) = 65 + 36 = 101$ ✓

$c = 13, d = 9$

(b) — instructor-judged. A correct response describes both routes and picks one. Matching the almond terms needs only the first equation doubled, because $2 \cdot 2 = 4$ already matches the second equation — one multiplication. Matching the cashew terms needs the first equation times 5 and the second times 3 (both become $15c$) — two multiplications and larger numbers. Full credit for naming the one-multiplication route and saying why it is fewer steps; partial credit for a correct route with no comparison.